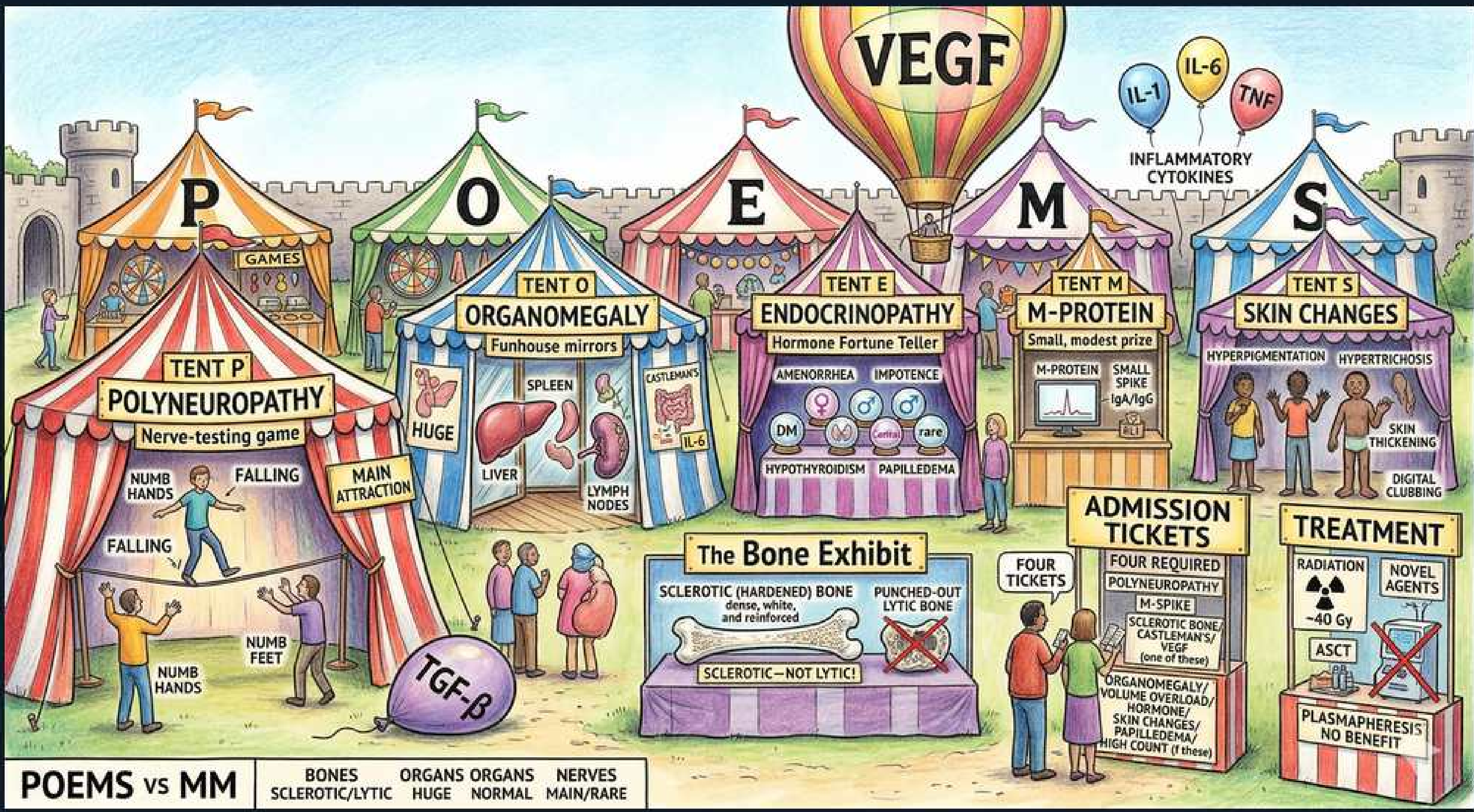
Plasma Cell — POEMS Syndrome



1. VEGF driver

WHAT YOU SEE

VEGF hot-air balloon over the fair

WHAT IT ENCODES

VEGF (vascular endothelial growth factor) is markedly elevated in POEMS and is the central pathogenic driver, causing microvascular hyperpermeability, angiogenesis, edema/effusions, organomegaly and skin changes. Serum/plasma VEGF correlates with disease activity and falls with successful treatment, making it the best marker for diagnosis and response. It is one of the major criteria. Memory hook: the VEGF balloon floats over the whole fairground because VEGF drives every other feature.

BOARD TRAP

WRONG answer: 'a normal VEGF rules out POEMS.' VEGF is the best biomarker and is elevated in the vast majority, but it is a MAJOR (not mandatory) criterion — diagnosis still requires the two mandatory features (polyneuropathy + monoclonal plasma-cell disorder). Discriminator: don't exclude POEMS solely on VEGF; conversely, mild VEGF elevation is nonspecific (also seen in other inflammatory states), so it supports but does not by itself establish the diagnosis.

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2. P = Polyneuropathy

WHAT YOU SEE

Tent P, numb hands/feet, falling

WHAT IT ENCODES

Chronic, progressive, symmetric, ascending sensorimotor peripheral neuropathy — the dominant and MANDATORY feature, present in essentially 100%. It is demyelinating with axonal loss on NCS/EMG, distal-predominant, and often mimics CIDP but is more painful and more motor-disabling. CSF shows albuminocytologic dissociation (high protein). Mnemonic: P is the first and biggest tent because neuropathy is usually the presenting complaint.

BOARD TRAP

WRONG answer: 'treat the neuropathy like CIDP with IVIG/steroids/plasma exchange.' POEMS neuropathy does NOT respond to immunomodulation aimed at the nerve — it improves only when the underlying plasma-cell clone is treated (radiation to a plasmacytoma, or systemic therapy ± ASCT). Discriminator from CIDP: POEMS has the monoclonal protein (usually lambda), high VEGF, sclerotic bone lesions, organomegaly, endocrinopathy — features absent in idiopathic CIDP.

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3. O = Organomegaly

WHAT YOU SEE

Tent O, huge liver/spleen/nodes

WHAT IT ENCODES

Hepatomegaly, splenomegaly, and lymphadenopathy — the nodes frequently show Castleman disease (angiofollicular hyperplasia) histology. Organomegaly is a MINOR criterion. It reflects the cytokine/VEGF-driven proliferative state rather than direct plasma-cell infiltration. Memory hook: the huge liver and spleen fill Tent O.

BOARD TRAP

WRONG answer: 'enlarged spleen + monoclonal protein = splenic marginal-zone or another lymphoma.' In POEMS the organomegaly is reactive/Castleman-associated, not malignant lymphomatous infiltration. Discriminator: look for the constellation (neuropathy, lambda M-protein, sclerotic bone, endocrinopathy, high VEGF) rather than attributing organomegaly to a primary lymphoma.

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4. E = Endocrinopathy

WHAT YOU SEE

Tent E fortune teller, amenorrhea/impotence/hypothyroid

WHAT IT ENCODES

Multiple endocrine axes are involved (a MINOR criterion): hypogonadism is the MOST common (amenorrhea, erectile dysfunction, gynecomastia), followed by hypothyroidism, glucose abnormalities/diabetes, and adrenal insufficiency. At least one endocrinopathy beyond diabetes/hypothyroidism is needed to count. Mnemonic: the fortune-teller in Tent E predicts a failing endocrine system across several glands.

BOARD TRAP

WRONG answer: 'the patient has type 2 diabetes and hypothyroidism, so the endocrine criterion is met.' Because diabetes and hypothyroidism are so common in the general population, they do NOT count on their own — you need involvement of another axis (gonadal, adrenal, parathyroid) to satisfy the endocrinopathy criterion. Discriminator: hypogonadism is the most specific/common axis in POEMS.

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5. M = Monoclonal protein

WHAT YOU SEE

Tent M, small monoclonal spike

WHAT IT ENCODES

A monoclonal plasma-cell disorder is MANDATORY and is almost always LAMBDA light-chain restricted (IgG-lambda or IgA-lambda). The M-spike is characteristically SMALL (often only detectable on immunofixation, not on SPEP), and marrow plasma cells are usually <10%. Mnemonic: Tent M displays only a tiny prize because the spike is small but its lambda type is diagnostic.

BOARD TRAP

WRONG answer: 'the M-spike is tiny and marrow plasma cells are <10%, so this is just MGUS / POEMS is excluded.' A small clone is the RULE in POEMS — do not exclude it for low burden. Discriminator: the lambda restriction plus neuropathy, high VEGF, and sclerotic bone lesions distinguish POEMS from benign MGUS; a kappa clone or large M-spike should make you reconsider the diagnosis.

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6. S = Skin changes

WHAT YOU SEE

Tent S, hyperpigmentation/hypertrichosis/clubbing

WHAT IT ENCODES

Cutaneous features (a MINOR criterion): diffuse HYPERpigmentation, HYPERtrichosis, skin thickening (scleroderma-like), digital clubbing, acrocyanosis/plethora, white nails, and glomeruloid hemangiomas (the most specific lesion). Driven by VEGF/cytokines. Mnemonic: Tent S shows dark, hairy, thickened, clubbed extremities.

BOARD TRAP

WRONG answer: 'hyperpigmentation + fatigue + low BP = primary adrenal insufficiency (Addison) alone.' In POEMS, glomeruloid hemangiomas and the broader POEMS constellation point to the syndrome; adrenal insufficiency can coexist as part of the endocrinopathy, but isolated Addison does not explain the neuropathy, M-protein, sclerotic bone and high VEGF. Discriminator: glomeruloid hemangioma is essentially pathognomonic for POEMS.

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7. Bone lesions sclerotic

WHAT YOU SEE

Bone exhibit, sclerotic hardened bone

WHAT IT ENCODES

POEMS bone lesions are osteoSCLEROTIC (dense, ivory-like) — the opposite of multiple myeloma's lytic lesions. They are a MAJOR criterion; a solitary sclerotic plasmacytoma is a common presentation and is curable with localized radiation. Best detected on plain films/CT (sclerotic) or PET. Mnemonic: the Bone Exhibit shows HARDENED, reinforced bone, not punched-out holes.

BOARD TRAP

WRONG answer: 'punched-out lytic lesions on skull/long bones favor POEMS.' Lytic, punched-out lesions are the hallmark of MULTIPLE MYELOMA, not POEMS — POEMS is sclerotic. Discriminator table: POEMS = sclerotic bone, normal calcium, normal renal function, normal/elevated platelets; Myeloma = lytic bone, hyperCalcemia, Renal failure, Anemia (CRAB).

8. Castleman / IL-6

WHAT YOU SEE

IL-1/IL-6/TNF inflammatory cytokines

WHAT IT ENCODES

POEMS overlaps with Castleman disease (a subset has coexisting multicentric Castleman). Pro-inflammatory cytokines — especially IL-6, plus IL-1 and TNF-alpha — are elevated alongside VEGF and contribute to organomegaly, effusions, thrombocytosis and constitutional symptoms. Castleman variant of POEMS can occur even without a detectable monoclonal protein. Mnemonic: the cytokine cloud (IL-1/IL-6/TNF) fuels the fair.

BOARD TRAP

WRONG answer: 'IL-6 elevation means anti-IL-6 (tocilizumab/siltuximab) is the primary therapy for POEMS.' IL-6 blockade targets Castleman physiology but is NOT the mainstay for classic POEMS — treatment is directed at the plasma-cell clone (radiation for localized disease, lenalidomide/melphalan-based therapy ± ASCT for systemic). Discriminator: anti-IL-6 has a role in Castleman-variant disease, not as standard POEMS therapy.

9. Major criteria pair

WHAT YOU SEE

Admission tickets: four required, polyneuropathy + monoclonal

WHAT IT ENCODES

Diagnosis requires BOTH mandatory criteria (polyneuropathy AND monoclonal plasma-cell disorder, almost always lambda) PLUS at least one of the other MAJOR criteria: osteosclerotic bone lesions, Castleman disease, or elevated VEGF. Then at least one MINOR criterion (organomegaly, endocrinopathy, skin changes, papilledema, extravascular volume overload, thrombocytosis/polycythemia) supports it. Mnemonic: the 'admission tickets' = 2 mandatory + 1 other major + ≥1 minor.

BOARD TRAP

WRONG answer: 'three of the five POEMS letters are enough to diagnose POEMS.' The acronym is a memory aid, not the diagnostic rule — you specifically NEED the two mandatory features plus an additional major criterion. Discriminator: a patient with organomegaly + endocrinopathy + skin changes but NO polyneuropathy and NO monoclonal protein does not have POEMS.

10. Papilledema / fluid

WHAT YOU SEE

Papilledema, volume overload, edema

WHAT IT ENCODES

Extravascular volume overload (a MINOR criterion) from VEGF-driven capillary leak: peripheral edema, ascites, pleural effusions, and pericardial effusion. PAPILLEDEMA is characteristic and is often asymptomatic — fundoscopy should be performed. Mnemonic: swelling everywhere, including the optic disc, because leaky vessels flood the tissues.

BOARD TRAP

WRONG answer: 'papilledema in this patient means idiopathic intracranial hypertension / a brain tumor — get urgent neuroimaging only.' In a patient with the POEMS constellation, papilledema is part of the syndrome (VEGF-mediated). Discriminator: pair the papilledema with neuropathy, M-protein, effusions and sclerotic bone; imaging is still reasonable but the unifying diagnosis is POEMS.

11. Thrombocytosis

WHAT YOU SEE

High platelet count note on tickets

WHAT IT ENCODES

THROMBOCYTOSIS and/or polycythemia (erythrocytosis) are MINOR criteria and are a classic POEMS clue — the OPPOSITE of myeloma's cytopenias. Driven by cytokines (IL-6, thrombopoietin effect). Useful at the bedside: a neuropathy patient with HIGH platelets and a monoclonal protein should trigger POEMS, not myeloma. Mnemonic: the platelet count is UP on the ticket board.

BOARD TRAP

WRONG answer: 'low platelets/anemia plus a monoclonal protein and bone disease = POEMS.' Cytopenias point AWAY from POEMS and TOWARD multiple myeloma (the 'A' anemia of CRAB). Discriminator: POEMS classically has elevated platelets and red cells; if the patient is cytopenic with lytic lesions and hypercalcemia, think myeloma.

12. Treatment radiation

WHAT YOU SEE

Treatment tent, radiation for isolated lesion

WHAT IT ENCODES

For LOCALIZED disease (1–3 osteosclerotic lesions without diffuse marrow involvement), targeted RADIOTHERAPY to the plasmacytoma(s) can be definitively curative, with neuropathy improving over months. This is the preferred first-line for limited disease. Mnemonic: aim the radiation beam at the isolated sclerotic lesion.

BOARD TRAP

WRONG answer: 'always start systemic chemotherapy/transplant regardless of extent.' For limited, localized osteosclerotic POEMS, RADIATION alone is first-line and can be curative — reserve systemic therapy for diffuse disease. Discriminator: assess marrow involvement and number of lesions (PET/marrow biopsy) before choosing radiation vs. systemic therapy.

13. Treatment ASCT

WHAT YOU SEE

ASCT in treatment tent

WHAT IT ENCODES

For WIDESPREAD/diffuse disease (diffuse marrow plasmacytosis or multiple lesions), use systemic therapy — lenalidomide-dexamethasone or melphalan-based induction — followed by high-dose melphalan and AUTOLOGOUS stem-cell transplant in eligible patients, which produces durable hematologic and neurologic responses. Mnemonic: the ASCT booth handles the 'spread-out' cases.

BOARD TRAP

WRONG answer: 'POEMS is incurable, so transplant offers no neurologic benefit.' ASCT produces substantial, durable improvement in the neuropathy and other features in eligible patients with disseminated disease. Discriminator: response is tracked by VEGF levels and neurologic exam, not just the M-protein (which is small).

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14. Plasmapheresis no benefit

WHAT YOU SEE

Plasmapheresis crossed out, no benefit

WHAT IT ENCODES

Plasmapheresis (plasma exchange) does NOT improve the neuropathy in POEMS — unlike other paraproteinemic neuropathies (e.g., anti-MAG IgM, or hyperviscosity in Waldenström). The neuropathy is VEGF/cytokine-driven, not directly from circulating M-protein, so removing the small M-protein does nothing. Mnemonic: the plasmapheresis machine is crossed out — 'NO BENEFIT.'

BOARD TRAP

WRONG answer: 'the patient has a paraproteinemic neuropathy, so plasma exchange should help.' Plasmapheresis helps anti-MAG IgM neuropathy and hyperviscosity, but is INEFFECTIVE in POEMS. Discriminator: in POEMS treat the clone (radiation or systemic therapy ± ASCT) — not the plasma; reaching for PLEX is the classic wrong move.
